MATHEMATICS OF COMPUTATION
Volume 90, Number 329, May 2021, Pages 1435–1453
https://doi.org/10.1090/mcom/3612
Article electronically published on February 26, 2021

LITTLEWOOD POLYNOMIALS, SPECTRAL-NULL CODES, AND EQUIPOWERFUL PARTITIONS

JOE BUHLER, SHAHAR GOLAN, ROB PRATT, AND STAN WAGON

ABSTRACT. Let $[n]$ denote $\{0, 1, ..., n-1\}$. A polynomial $f(x) = \sum a_i x^i$ is a *Littlewood polynomial* (LP) of length n if the a_i are ± 1 for $i \in [n]$, and $a_i = 0$ for $i \geq n$. An LP has *order* m if it is divisible by $(x-1)^m$. The problem of finding the set L_m of lengths of LPs of order m is equivalent to finding the lengths of spectral-null codes of order m, and to finding n such that $[n]$ admits a partition into two subsets whose first m moments are equal. Extending the techniques and results initiated by Boyd, we completely determine L_7 and L_8, and prove that $\min L_9 = 192$ and $\min L_{10} = 240$. Our primary tools are (a) symmetry, and (b) the use of carefully targeted searches using integer linear programming both to find LPs and to disprove their existence for specific n and m. Symmetry plays an unexpected role and leads to the concept of *regenerative pairs*, which produce infinite arithmetic progressions in L_m. We prove for $m \leq 8$ (and conjecture later for all m) that whenever there is an LP of length n and order m, there is one of length n and order m that is symmetric (resp. antisymmetric) if m is even (resp. odd).

1. INTRODUCTION

The partition of $\{0, 1, 2, 3, 4, 5, 6, 7\}$ into the two disjoint sets $A = \{0, 3, 5, 6\}$ and $B = \{1, 2, 4, 7\}$ is especially well-balanced ("equipowerful") in that the first three moments of A are equal to the corresponding moments of B; i.e., $\sum_{a \in A} a^j = \sum_{b \in B} b^j = \frac{1}{2} \sum_{i=0}^{7} i^j$ for $j = 0, 1, 2$. Figure 1 illustrates the three identities geometrically. The bipartition has *length* 8, and is 3-*equipowerful* (or has *order* 3).

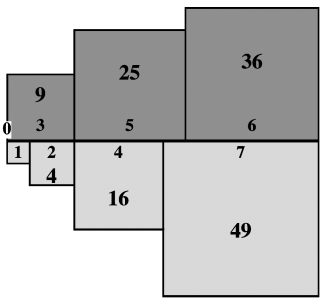


FIGURE 1. A 3-equipowerful bipartition of length 8; the three power-sums are 4, 14, 70.

Received by the editor December 9, 2019, and, in revised form, August 24, 2020, and September 29, 2020.

2020 *Mathematics Subject Classification.* Primary 11B83, 12D10; Secondary 94B05, 11Y99.

Key words and phrases. Littlewood polynomials, spectral-null code, antenna array, multigrade identity, equal power sum partition, integer linear programming.

©2021 American Mathematical Society

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

A bipartition of this kind can be encoded algebraically as a generating function in two natural ways: either as a Littlewood polynomial (all coefficients ± 1) via $f(x) = \sum_{a\in A} x^a - \sum_{b\in B} x^b$, or as a trigonometric polynomial $g(t) = f(\mathbb{e}^{\mathbb{i}t}) = \sum_{a\in A} \mathbb{e}^{\mathbb{i}at} - \sum_{b\in B} \mathbb{e}^{\mathbb{i}bt}$, all of whose coefficients are ± 1. The fact that the preceding bipartition is 3-equipowerful is equivalent to $f(x)$ vanishing to order 3 at $x = 1$, and also to $g(t)$ vanishing in t to order 3 at $t = 0$.

Littlewood polynomials have appeared in several contexts [2,7], and polynomials with $\{-1, 0, 1\}$ coefficients, or even more general constraints [5,11], have been considered. Trigonometric polynomials with $\{\pm 1\}$ coefficients correspond to "spectral-null codes" [14,16] which arise in signal processing in several important ways, e.g., in encoding digital information on media (such as a DVD) where low frequencies need to be suppressed. Applications for small m use efficient algorithms for encoding arbitrary bit strings into (the somewhat longer) strings in some spectral-null code of order m. Using codes for larger m will require reasonably efficient encoding/decoding algorithms, and a better understanding of what lengths are possible for codes of some specific order m.

This question, or equivalent versions in other guises, is the central focus of this paper: given a positive integer m, what are the possible lengths of (equivalently) an equipowerful bipartition, a Littlewood polynomial, or a spectral-null code, of order m? This set will be written

$$L_m = \{n : \text{there is an LP of length } n \text{ and order } m\}.$$

This has been investigated in the Littlewood polynomial context by, among others, Boyd [6,7], Berend and Golan [10], and Freiman and Litsyn [9].

The primary tools that we use to significantly extend known results are (1) carefully targeted searches that use integer linear programming (ILP), and (2) an unexpected concept that we call "regenerative pairs," which yields efficient proofs of the existence of infinite families of order-m LPs and suggests additional structure for LPs of high order.

Section 2 gives the required definitions, §3 states our results, §4 gives the background theorems that are needed, §5 has proofs of the negative results, and §6 gives positive results obtained by ILP and introduces the concept of regenerative pairs, which are used to find infinite families of LPs. The final section states some conjectures that emerge from the data. Our work shows how one can use experiments, aided by sophisticated computations (ILP) to generate hunches based on patterns, which can be used to refine the experiments and also as a guide to proofs. One danger of course is that the patterns are not as predictive as one might hope and we enumerate a series of counterexamples to natural conjectures, including a spectacular one that has order 52 and length 2^{51}.

2. Notation and definitions

We use $[n]$ to denote $\{0, 1, \ldots, n-1\}$. A decomposition $[n] = A \cup B$ into two disjoint subsets is called *m-equipowerful*, or *of order m*, if the first m moments (power-sums) of the sets are equal, i.e., $\sum_{a\in A} a^j = \sum_{b\in B} b^j$ for $j = 0, 1, \ldots, m-1$. We write $A \stackrel{m}{=} B$ when the first m moments of A and B are equal; m is called the *order* of the bipartition and A and B are called *witnesses*. We always assume that m is positive, so that the two subsets have the same size. The equations $A \stackrel{m}{=} B$ are

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

an example of a "multigrade identity" and also an instance of the general Prouhet–Tarry–Escott problem [4, 8].

A *Littlewood polynomial* (LP) is a polynomial whose exponents are $[n]$ for some n and whose coefficients are all ± 1. The degree of a polynomial f is denoted $\deg(f)$; the *length* of a polynomial f, denoted $\ell(f)$, is the number of its coefficients, i.e., $\ell(f) = \deg(f) + 1$. An LP $f(x)$ has *order* m if $(x-1)^m$ divides $f(x)$. We will have occasion to refer to maximal such m, and define f to have *exact order* m if it is divisible by $(x-1)^m$, but not by $(x-1)^{m+1}$.

As will be shown in §4, equipowerful partitions of $[n]$ of order m and LPs of length n and order m are equivalent concepts. Since m is positive, the length n is always even. The ideas extend immediately to bipartitions of any interval I of n consecutive integers (or to LPs with nonzero coefficients in I). Multiplying an LP by a power of x does not change the power of $x-1$ that divides it, so it suffices to consider polynomials with $I = [n]$ and, by the aforementioned equivalence (or a direct proof), it suffices to consider bipartitions of $[n]$.

We let $\mathcal{P}(n, m)$ denote the set of LPs of length n and order m, and L_m the set of lengths of LPs of order m: $L_m = \{n : \mathcal{P}(n, m) \neq \varnothing\}$. The initial goal of this paper is to discover as much as possible about L_m, for small m. The behavior of the smallest element of L_m, as a function of m, will be a focus of interest.

A dual concept is also useful: let $m^*(n)$ be the largest m so that $n \in L_m$. For example, $m^*(8) = 3$ because the example of §1 gives $1 - x - x^2 + x^3 - x^4 + x^5 + x^6 - x^7 = (1-x)^3(1+x)^2(1+x^2)$, which is in $\mathcal{P}(8, 3)$, and an easy search shows that $\mathcal{P}(8, 4)$ is empty.

The *join* of LPs $f(x)$ and $g(x)$ is $(f \vee g)(x) = f(x) + x^{\ell(f)}g(x)$. The *expanded product* of LPs f and g, denoted $f\#g$, is $(f\#g)(x) = f(x)g(x^{\ell(f)})$, which is an LP; this product is associative but not commutative.

Symmetry plays a large role in our investigations, so we introduce several concepts related to symmetric and antisymmetric LPs. The *reversal* of an LP $f(x)$ is $f^*(x) = x^{\deg(f)}f(1/x)$ and f is *symmetric* (or *+symmetric*) if $f^* = f$ and *antisymmetric* (or *−symmetric*) if $f^* = -f$. If $s = 1$ or $s = -1$, then the *s-symmetrization* (or *symmetrization* if $s = 1$ and *antisymmetrization* if $s = -1$) of an LP f is $f \vee (sf^*)$.

Finally, we introduce a special polynomial that is a seminal example in our story.

Definition 2.1. The *Thue–Morse polynomial*, which we denote τ_m, is

$$\tau_m(x) = (1-x)(1-x^2)\dots\left(1-x^{2^{m-1}}\right).$$

The polynomial τ_m has length 2^m and is $(-1)^m$ symmetric. These polynomials played an important role in the early work because τ_m has order m, which means $\tau_m \in \mathcal{P}(2^m, m)$, $2^m \in L_m$, and $m^*(2^m) \geq m$. We note that $\tau_m(x) = (1-x)\#(1-x)\#\cdots\#(1-x)$ where the expanded product is used $m-1$ times.

In our ILPs the variables are x_i, $i \in [n]$, with each x_i constrained to be 0 or 1. The objective function is always 0, so that we are determining feasibility, as opposed to optimizing a nonconstant function. We use two types of ILP. For the basic search (§6), we seek $A \subseteq [n]$ giving an equipowerful partition: $x_i = 1$ means that $i \in A$. The constraints are those arising from the power-sum equations; for exponent j, the constraint is the linear equation $\sum_{i=0}^{n-1} i^j x_i = \frac{1}{2}\sum_{i=0}^{n-1} i^j$. We often include constraints based on symmetry or antisymmetry. A variation of this is used to find regenerative pair examples (§6); constraints are added for the smaller of

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

the pair. The second type (§5) searches for $A \subseteq [n]$ with specified residue counts modulo primes. The linear constraints are defined to force the desired counts. The general approach is to work up through the primes, usually including residue-count constraints for the primes smaller than p. Auxiliary integer variables are used to implement the divisibility conditions that arise in Proposition 4.2 and 4.3.

3. New results

Work of Boyd [6, 7] and Berend and Golan [2] gave exact descriptions of L_m for $m \leq 6$, and also exact values of $m^*(n)$ for $n \leq 167$ and $n = 256$; these are included in the tables below.

Their results for L_m are extended here to exact descriptions for $m = 7$ and $m = 8$, as well as a determination of the elements of L_9 that are smaller than 304. The extensive and carefully structured computations required to do this will be described in §5 and §6. The next theorem states the new results precisely, and the full story is given in Table 1. We let $\mathbb{N}$ denote the set of nonnegative integers.

Theorem 3.1. *The set L_7 contains all multiples of 16 from 96 onward, and all multiples of 8 from 192 onward. The set L_8 contains 144 and all multiples of 16 from 192 onward. More succinctly, $L_7 = \{96, 112, 128, 144, 160, 176\} \cup (192 + 8\,\mathbb{N})$ and $L_8 = \{144\} \cup (192 + 16\,\mathbb{N})$. The only integers in L_9 that are less than 304 are 192, 240, and 288. Moreover, 240 is in L_{10}.*

Table 1. L_m, for $m \leq 8$.

m	L_m
1	$2 + 2\,\mathbb{N}$
2	$4 + 4\,\mathbb{N}$
3	$8 + 4\,\mathbb{N}$
4	$16 + 8\,\mathbb{N}$
5	$32 + 8\,\mathbb{N}$
6	$\{48\} \cup (64 + 8\,\mathbb{N})$
7	$\{96, 112, 128, 144, 160, 176\} \cup (192 + 8\,\mathbb{N})$
8	$\{144\} \cup (192 + 16\,\mathbb{N})$

Note that in Table 1 the eventual differences of the sets L_m are 2, 4, 4, 8, 8, 8, 8, 16. In fact, the Divisibility condition in Theorem 4.1 below says if $\mathcal{P}(n, m) \neq \varnothing$ then n is divisible by the smallest power of 2 that is strictly bigger than m. For example, any number in L_2 or L_3 has to be divisible by 4; and L_4 through L_7 are contained in $8\,\mathbb{N}$. Remarkably, this divisibility criterion is not only necessary for membership in L_m, but is almost sufficient. Freiman and Litsyn [9, Thm. 1] prove that, roughly speaking, the chance that a random LP of length n is divisible by $(x-1)^m$ is about $n^{-m^2/2}$, if n satisfies the necessary divisibility condition. More precisely, $|\mathcal{P}(n, m)| = c\,(1 + o(1))\,2^n n^{-m^2/2}$, where the constant c is explicit and depends only on m, and the $o(1)$ term goes to zero as n goes to infinity. Thus if n is large enough and divisible by the requisite power of 2, then $\mathcal{P}(n, m)$ is nonempty. Ignoring the $o(1)$ term, and using the explicit formula for c, turns out to suggest that $\mathcal{P}(n, m)$ is nonempty once n is at least $O(m^2 \log(m))$. It is known, as explained in [5], that the smallest degree of a polynomial divisible by $(x-1)^m$

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

with coefficients in $\{-1, 0, 1\}$ is, asymptotically, between a constant multiple of m^2 and a constant multiple of $m^2 \log(m)$. However, this is definitely not true for Littlewood polynomials. Their greater rigidity, as exemplified by the behavior of their values at roots of unity (glimpsed below in §4), enabled Boyd [6] to prove that $\min L_m \geq \exp(\sqrt{m}(1 + o(1)))$. Thus, asymptotically, the smallest element of L_m lies between $\exp(\sqrt{m}(1 + o(1)))$ and, from the Thue–Morse polynomial, 2^m.

Our work reveals further structure in equipowerful sets. Anyone working in the area is drawn to the ubiquitous symmetry properties of the witnessing sets and polynomials. For instance, it appears that whenever $n \in L_m$ there is a witnessing set that is $(-1)^m$ symmetric. This is easy to prove when $m = 2$: just apply symmetry to extend all the initial segments of the infinite sequence $(0, 1, 2, 3, \dots)$, getting $\{0, 3\}$, $\{0, 1, 6, 7\}$, $\{0, 1, 2, 9, 10, 11\}$, and so on. Yet this had not been proved even for $m = 3$. However, we now know that $(-1)^m$ symmetric witnesses exist for all entries in Table 1.

Theorem 3.2. *In all the cases of Table 1, there is a $(-1)^m$ symmetric LP in $\mathcal{P}(n, m)$.*

Recall that $m^*(n) = \max\{m : \mathcal{P}(n, m) \neq \varnothing\}$, i.e., $m = m^*(n)$ is the largest power of $x - 1$ that divides some LP of length n. It follows from Table 1 that $m^*(2k+1) = 0$, $m^*(4k+2) = 1$, and $m^*(8k+4) = 2$ for nonnegative k, so we next consider n divisible by 8. From earlier work, $m^*(n)$ was known for all $n = 8k$ for which $m^*(n) \leq 7$, as well as $m^*(144) = m^*(256) = 8$ (the 256 value is in [2]). Our calculations found the following new values.

Theorem 3.3. *$m^*(168) = m^*(184) = 6$, $m^*(176) = m^*(200) = m^*(216) = m^*(232) = m^*(248) = m^*(264) = m^*(280) = m^*(296) = 7$, $m^*(208) = m^*(224) = m^*(272) = 8$, $m^*(192) = 9$, $m^*(240) \leq 11$, $m^*(288) \leq 10$, and $m^*(304) \leq 9$.*

Table 2 combines these results with what was known from earlier work. Note that $n = 240$ is the smallest value for which $m^*(n)$ is not known exactly. The remarkable 2^{51} result is due to Richard Stong, as will be described in detail later (Theorem 7.1).

TABLE 2. Values of $m^*(n)$.

n	8	16	24	32	40	48	56	64	72	80	88	96	104	112	120	128
$m^*(n)$	3	4	4	5	5	6	5	6	6	6	6	7	6	7	6	7

n	136	144	152	160	168	176	184	192	200	208	216	224	232
$m^*(n)$	6	8	6	7	6	7	6	9	7	8	7	8	7

n	240	248	256	264	272	280	288	296	304	2^{51}
$m^*(n)$	10 or 11	7	8	7	8	7	9 or 10	7	8 or 9	≥ 52

4. Basic theorems

We start with several classic results about equipowerful sets. The first gives the important equivalence between LPs, equipowerful sets, and spectral-null codes.

Equivalence Lemma. *For a bipartition A, B of $[n]$ and an integer $m \geq 1$, the following are equivalent.*

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

(1) $A \stackrel{m}{=} B$.
(2) t^m *divides the Maclaurin series of* $g(t) = \sum_{a\in A} \mathbb{e}^{\mathbb{i}at} - \sum_{b\in B} \mathbb{e}^{\mathbb{i}bt}$.
(3) $(x-1)^m$ *divides the polynomial* $f(x) = \sum_{a\in A} x^a - \sum_{b\in B} x^b$.

Proof. $1 \Leftrightarrow 2$. Let $(a_j)_{0\le j<n}$ be the ± 1 sequence of length n that corresponds to A, B, so that $g(t) = \sum_j a_j \mathbb{e}^{\mathbb{i}jt}$. Then (2) holds if and only if the first m derivatives $g^{(k)}(0)$, $0 \le k \le m-1$, all equal 0. But $g^{(k)}(0) = \mathbb{i}^k \sum_{j=0}^{n-1} a_j j^k$, so the vanishing of the derivatives is equivalent to $A \stackrel{m}{=} B$.

$2 \Leftrightarrow 3$. An easy induction argument, using the chain and product rules, shows that there are integers $s_{j,k}$ such that

$$g^{(k)}(t) = f^{(k)}(x)(\mathbb{i}x)^k + \mathbb{i}^k \sum_{j=1}^{k-1} s_{j,k}\, f^{(j)}(x)\, x^j,$$

where $x = \mathbb{e}^{\mathbb{i}t}$. If all of the jth derivatives $g^{(j)}(0)$ and $f^{(j)}(1)$ vanish for $j \le k-1$, then it follows that $g^{(k)}(0) = 0$ and $f^{(k)}(1) = 0$ are equivalent, and therefore g has order m at 0 if and only if f has order m at 1. □

The next theorem collects various known results. The Multiplication result is in [2, Prop. 2.4]. Recall that the join and expanded product operators are defined by $(f \vee g)(x) = f(x) + x^{\ell(f)} g(x)$ and $(f\#g)(x) = f(x)g(x^{\ell(f)})$ respectively.

Theorem 4.1.

Thue–Morse: *$\tau_m \in \mathcal{P}(2^m, m)$, and therefore $2^m \in L_m$.*
Addition: *If $f \in \mathcal{P}(n_1, m)$ and $g \in \mathcal{P}(n_2, m)$ then $f \vee g \in \mathcal{P}(n_1 + n_2, m)$.*
Doubling: *If $f \in \mathcal{P}(n, m)$ then $f \vee (-f) \in \mathcal{P}(2n, m+1)$.*
Multiplication: *If $f \in \mathcal{P}(n, m)$ and $g \in \mathcal{P}(n', m')$, then $f\#g \in \mathcal{P}(n\,n', m+m')$.*
Symmetry: *If f is a $(-1)^m$ symmetric LP that is divisible by $(x-1)^{m-1}$ then f is divisible by $(x-1)^m$.*
Divisibility: *Let $n = 2^k u$ where u is odd. If $\mathcal{P}(n, m)$ is nonempty then $2^k > m$.*

Proof. Thue–Morse. Since $\tau_{m+1} = \tau_m \vee -\tau_m$ it follows that τ_m is an LP, and has length 2^m and order m, by induction. Note also that $\tau_{m+1} = \tau_m \# (1-x) = (1-x)\#\tau_m$.

Addition. The join $f \vee g$ is obviously an LP, has length $n_1 + n_2$ and is divisible by $(x-1)^m$, as asserted. Note that this implies that L_m is closed under addition.

Doubling. If f is in $\mathcal{P}(n, m)$ then $g = (1-x^n)f(x) = f \vee (-f)$ has length $2n$ and is divisible by $(1-x)^{m+1}$ since $1 - x^n$ introduces an extra $1 - x$ factor. In other words, if $n \in L_m$ then $2n \in L_{m+1}$.

Multiplication. The polynomial $h(x) = f(x)g(x^n)$ consists of n' blocks of length n, each the result of multiplying the coefficients of f by a single coefficient of g. In particular h is an LP. The polynomial h is divisible by $(x-1)^m (x^n - 1)^{m'}$ and therefore divisible by $(x-1)^{m+m'}$. Note that Doubling is the special case $g(x) = 1 - x$ of Multiplication.

Symmetry. Substituting $f(x) = (x-1)^{m-1} g(x)$ into $f(x) = (-1)^m f^*(x)$ gives

$$(x-1)^{m-1} g(x) = (-1)^m x^{m-1+\deg(g)} (1/x - 1)^{m-1} g(1/x) = -(x-1)^{m-1} g^*(x).$$

Dividing by $(1-x)^{m-1}$ gives $g(x) = -g^*(x)$, which implies that $g(1) = g^*(1) = 0$ so that $g(x)$ is divisible by $x - 1$, from which the result follows. Note that this

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

implies that if f has exact order m, and is either symmetric or antisymmetric, then f is $(-1)^m$ symmetric.

Divisibility. The key point is that the number of factors of $1-x$ in $f(x)$ modulo 2 (i.e., working in the ring of polynomials over the 2-element field $\{0,1\}$) is at least as large as the number of such factors when f is thought of as a polynomial with integer coefficients.

Suppose that $f \in \mathcal{P}(n,m)$. Then $f(x) \equiv 1 + x + x^2 + \cdots + x^{n-1} \pmod 2$ and $(1-x)f(x) \equiv 1 - x^n$, where $\equiv$ will denote coefficient-wise congruence modulo 2. Note that $1 - x^2 \equiv (1-x)^2$. Iterating this shows that $1 - x^t \equiv (1-x)^t$, where $t = 2^k$. Introducing the shorthand $y = x^t$ gives

$$(1-x)f(x) \equiv 1 - x^n = 1 - x^{tu} = (1-y)\left(1 + y + \cdots + y^{u-1}\right) \equiv (1-x)^t g(x)$$

where $g(1) = u \equiv 1$ so that $g(x)$ is not divisible by $1-x$. Counting the factors of $1-x$ gives $1 + m \le 2^k$ as claimed. □

Remarks.

1. Doubling reverses symmetry in the sense that it turns a symmetric polynomial into an antisymmetric one, and vice versa. Addition does not preserve either symmetry.

2. Thue–Morse, Addition, and Doubling imply that

$$m^*(2^m) \ge m, \quad m^*(n_1 + n_2) \ge \min\left[m^*(n_1), m^*(n_2)\right], \text{ and } m^*(2n) \ge m^*(n) + 1.$$

3. The sets arising from the τ_k are initial segments of the infinite Thue–Morse sequence $0, 3, 5, 6,\ 9, 10, 12, 15, 17, 18, 20, 23, 24, 27, 29, 30, \ldots$. This sequence is exactly the set of so-called *evil numbers*: integers with an even number of 1s in their binary expansion. Note also that this sequence has one entry in common with each pair $\{0,1\}, \{2,3\}, \{4,5\}, \ldots$. Our heuristic searches often tried to prioritize this feature (or similar Thue–Morse-like properties) to aim for earlier success.

4. Divisibility says that if $\mathcal{P}(n,m) \ne \varnothing$, then n has to be divisible by the smallest power of 2 that is strictly bigger than m. As mentioned earlier, the main result of [9] implies that this necessary condition is also sufficient for sufficiently large m.

5. All of these results, except Divisibility, apply to more general polynomials.

We now find L_m for small m. The set L_1 consists of all n such that there is some length-n LP f that is divisible by $x-1$; this is the same as saying that $f(1) = 0$. Such an LP exists if and only if n is even, so L_1 is the set of positive even integers, which we write as $2 + 2\mathbb{N}$.

By Divisibility, $L_2 \subseteq 4 + 4\mathbb{N}$, and $\tau_2 = 1 - x - x^2 + x^3 \in \mathcal{P}(4,2)$ so $4 \in L_2$. Repeatedly joining τ_2, and using Addition, one shows that $L_2 = 4 + 4\mathbb{N}$.

Although $L_3 \subseteq 4 + 4\mathbb{N}$ by Divisibility, an easy search shows that no LP of length 4 is divisible by $(x-1)^3$. Because $\tau_3 \in \mathcal{P}(8,3)$ we know that $8 + 8\mathbb{N} \subseteq L_3 \subseteq 8 + 4\mathbb{N}$. A computer (or even hand) search finds that

$$f = 1 - x + x^2 - x^3 - x^4 - x^5 + x^6 + x^7 + x^8 - x^9 + x^{10} - x^{11} \equiv 0 \bmod (x-1)^3.$$

By the Equivalence Lemma this is equivalent to the fact that $\{0, 2, 6, 7, 8, 10\}$ is a 3-equipowerful subset of [12]. By repeatedly joining τ_3 onto τ_3 or f (i.e., using Addition) it follows that $L_3 = 8 + 4\mathbb{N}$. This characterization is due to Boyd [6].

Similar but more elaborate calculations (done in [2]) show that $L_4 = 16 + 8\mathbb{N}$ and $L_5 = 32 + 8\mathbb{N}$. The natural guess that $L_6 = 64 + 8\mathbb{N}$ turns out to be wrong, as discovered by Boyd [6], because there is a (unique up to sign) polynomial in

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

$\mathcal{P}(48, 6)$ (i.e., 48 is 6-equipowerful); he also proved [7] that 30, 40, and 56 are not in the set; Berend and Golan [2] proved that $L_6 = \{48\} \cup (64 + 8\mathbb{N})$. Theorem 3.1 specifies L_m exactly for $m = 7$ and $m = 8$.

A key to understanding the structure of the LPs in $\mathcal{P}(n, m)$ is generating various constraints that the polynomials and the corresponding equipowerful sets must satisfy. The Divisibility condition in Theorem 4.1 is an example of such a constraint. But there are many more and any computer search for equipowerful sets beyond modest values of n requires the use of a wide variety of constraints.

Here we use the following notations. For a set X and prime p, $C_{p,j}(X)$ denotes the number of elements of X that are congruent to $j \bmod p$, namely $C_{p,j}(X) = |X \cap (p\mathbb{Z} + j)|$. For an m-equipowerful bipartition of $[n]$ into A, B, put $d_{p,j} = C_{p,j}(A) - C_{p,j}(B)$. And we use $\boldsymbol{d}_p$ for the vector of $d_{p,j}$ values, $j = 0, 1, \ldots, p-1$.

The preceding definition formalizes the idea of the discrepancy between the sets in a witnessing bipartition. If $\boldsymbol{d}_2 = (a, b)$, then the even numbers in A exceed those in B by a, while the odds in A exceed the odds in B by b. So if $\boldsymbol{d}_2 = (0, 0)$, then the even and odd counts are the same for A and B; in this case each set would have $n/4$ evens and $n/4$ odds. This uniformity of parity happens for the sets arising from the Thue–Morse polynomials.

The next lemma gives several basic constraints regarding the value of $d_{p,j}$.

Proposition 4.2. *Suppose $d_{p,j}$ are defined from an m-equipowerful bipartition $A \cup B$ of $[n]$. Then:*

(1) $|d_{p,j}| \le C_{p,j}([n]) \le \left\lceil \frac{n}{p} \right\rceil$.
(2) $\sum_{j=0}^{p-1} d_{p,j} = 0$.
(3) *For each j, $d_{p,j} - C_{p,j}([n])$ is even.*

Proof. (1) holds because the extreme case has all the $j \bmod p$ numbers in one of the sets. (2) reduces to $|A| = |B|$. For (3), $d_{p,j} = C_{p,j}(A) - C_{p,j}(B) \equiv C_{p,j}(A) + C_{p,j}(B) = C_{p,j}([n]) \pmod 2$. □

The following propositions, with the same assumptions about m, n, A, B, provide constraints on $d_{p,j}$ that depend on m and the prime p. The results are immediate consequences of items 3.3 and 3.4 in [2]. The proof method in [2] builds upon ideas introduced by Boyd that involve substituting roots of unity for x and then using facts about the cyclotomic fields generated by those roots of unity.

Proposition 4.3. *Suppose $m = s(p-1) + r$, with $0 \le r \le p-2$. Then:*

(1) *If $r = 0$, then $d_{p,0}, d_{p,1}, \ldots, d_{p,p-1}$ are all congruent modulo p^s.*
(2) *If $r \ge 1$, then each $d_{p,j}$ is divisible by p^s and, for $0 \le j \le r-1$,*

$$\binom{j}{j} d_{p,j} + \binom{j+1}{j} d_{p,j+1} + \cdots + \binom{p-1}{j} d_{p,p-1} \equiv 0 \pmod{p^{s+1}}.$$

The next result gives a specific constraint on $\boldsymbol{d}_2$, asserting that under certain conditions the partition must be very unbalanced in terms of parity.

Proposition 4.4. *If $m = 2^q - 1$ and 2^q divides n exactly, then $|d_{2,0} - d_{2,1}| \ge 2^{2^q - 1}$.*

This result is proved in [2] using a result from [12]. We will give a self-contained proof here. First, an easy lemma from [7].

Lemma 4.5. *If $f \in \mathcal{P}(n, m)$ and $f(-1) \ne 0$, then $|f(-1)| \ge 2^m$.*

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

Proof. Write $f(x) = (x-1)^m g(x)$. Then $0 < |f(-1)| = 2^m\,|g(-1)|$, where $g(-1) \in \mathbb{Z}$. Therefore $|f(-1)| \geq 2^m$. □

The next proposition is due to Berend and Golan [2] and extends [7, Cor. 1].

Proposition 4.6. *If $n = tu$, with t a power of 2 and u odd, and f is an element of $\mathcal{P}(n,m)$ that is divisible by $x+1$, then $m < t-1$.*

Proof. We are given that $f(x) = (x-1)^m(x+1)h(x)$ for some polynomial h. From the proof of Divisibility above, we know that $x^n - 1 \equiv (x-1)^t g(x) \pmod 2$, where $g(x)$ is not divisible by $x-1$ modulo 2. The result follows immediately:

$$x^n - 1 \equiv (x-1)f(x) = (x-1)^{m+1}(x+1)h(x) \equiv (x-1)^{m+2}h(x) \equiv (x-1)^t g(x)$$

so that $m+2 \leq t$, or $m < t-1$, as claimed. □

Proposition 4.4 then follows from Lemma 4.5 and Proposition 4.6, the Equivalence Lemma, and the fact that $|d_{2,0} - d_{2,1}| = |f(-1)|$, where f is the LP defined from the equipowerful set A.

5. Nonexistence of equipowerful bipartitions

Here we show how the constraints of §4 yield negative results. Theorem 5.1, together with the positive results in §6, will prove Theorems 3.1, 3.2, and 3.3.

Theorem 5.1. $168, 184 \notin L_7$; $176, 200, 216, 232 \notin L_8$; $208, 224, 272 \notin L_9$; $192, 304 \notin L_{10}$; $288 \notin L_{11}$; *and* $240 \notin L_{12}$.

Proof. Because 16 does not divide 200, 216, or 232, these follow from Divisibility. The cases of 208 and 224, with $m = 9$, are handled as follows. Let $p = 3$. Then $s = 4$, $r = 1$ in Proposition 4.3, and by Propositions 4.2 and 4.3, $|d_{3,1}| \leq 75$ and $d_{3,1}$ is divisible by 81. Therefore $d_{3,1} = 0$, contradicting the fact that $C_{3,1}([n])$ is odd. Similarly, for 304 with $m = 10$, by Propositions 4.2 and 4.3 all $d_{3,i}$s are divisible by 81 and congruent modulo 243; but $d_{3,0}$ and $d_{3,2}$ have different parity and are bounded by 102, which leads to a contradiction. For 192 and $m = 10$, consider $p = 5$; then $s = r = 2$. By Propositions 4.2 and 4.3(1), $\boldsymbol{d}_5$ must be $(\pm 25, \mp 25, 0, 0, 0)$. But then the $j = 1$ case of Proposition 4.3(2) looks at the sum of $(\mp 25, 0, 0, 0)$, which is not divisible by p^{s+1}, or 125.

For $240 \notin L_{12}$, we first use $p = 7$. Then $s = 2$ and $r = 0$; by Proposition 4.2, $|d_{7,0}| \leq \lceil n/p \rceil = 35$. The mod-49 congruence conditions then force $\boldsymbol{d}_7 = (\pm 35, \pm 35, \mp 14, \mp 14, \mp 14, \mp 14, \mp 14)$ and we may use symmetry to force $d_{7,0} = d_{7,1} = 35$, meaning all 70 elements of the first two mod-7 classes are in the witnessing set. There are 1100 $\boldsymbol{d}_{11}$ vectors and (with one exception) each of them, viewed as residue counts, contains at least one 0; because every congruence class modulo 11 intersects $7\mathbb{Z} \cap [240]$, this gives a contradiction. The exception is the residue-count vector $(11, 11, 11, 11, 22, 11, 11, 11, 11, 5, 5)$. The first two mod-7 classes have an intersection of 6 elements with $(11\mathbb{Z} + 10) \cap [240]$, which is a contradiction.

The case of 288 is similar to 240 because $\boldsymbol{d}_5 = (50, 0, -50, 25, -25)$ and so the mod-5 residue counts are $(54, 29, 4, 41, 16)$. Similarly, the mod-7 residue counts are

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

one of

$$(0,24,24,24,24,24,24) \quad (21,17,38,10,31,3,24)$$
$$(21,38,31,24,17,10,3) \quad (21,10,24,38,3,17,31)$$
$$(21,31,17,3,38,24,10) \quad (21,3,10,17,24,31,38)$$
$$(21,24,3,31,10,38,17) \quad (42,17,17,17,17,17,17).$$

All but the last have a 0 or 3, which leads to a contradiction by a counting argument as before. For the last one, the mod-5 counts of multiples of 7 are 9,8,9,8,8 and so there are only 33 not congruent to 2 (mod 5). Adding 4 for the 2-class in $(54,29,4,4,26)$ gives 37, less than the required 42.

The four remaining cases are more difficult, but yield to a detailed computer-aided analysis of the constraints for small primes.

$n = 184$, $m = 7$. Assume that A, B witness $184 \in L_7$; so $|A| = |B| = 92$. We will consider the primes $p \le 13$, learning all possibilities for $\boldsymbol{d}_p$ in each case, taking into account previous cases as we move up. Then at $p = 17$ there will be no possibility for $d_{17,16}$ consistent with the results for smaller primes.

We start with $p = 2$; then $s = 7$ and $r = 0$. We have

- $|d_{2,0}| \le 92$ by Proposition 4.2;
- $d_{2,1} = -d_{2,0}$, by Proposition 4.2;
- 128 divides $d_{2,1} - d_{2,0}$, by Proposition 4.3;
- $|d_{2,1} - d_{2,0}| \ge 128$, by Proposition 4.4.

The first three mean that $\boldsymbol{d}_2$ is $(64,-64)$, $(0,0)$, or $(-64,64)$. The last item eliminates $(0,0)$. Switching A and B if necessary, we can assume that $d_{2,0} \ge 0$. This proves $\boldsymbol{d}_2 = (64,-64)$, which means that the even-odd distribution in the witnessing sets is $(78,14)$ for A, and $(14,78)$ for B. We can invoke this switching trick once only.

Now let $p = 3$; then $s = 3$ and $r = 1$. We have

- $|d_{3,j}| \le 62$, by Proposition 4.2;
- $d_{3,0} + d_{3,1} + d_{3,2} = 0$, by Proposition 4.2;
- $d_{3,0}$ is even and $d_{3,1}$, $d_{3,2}$ are odd by Proposition 4.2, because $(C_{3,0}, C_{3,1}, C_{3,2}) = (62,61,61)$;
- 27 divides each $d_{3,j}$, by Proposition 4.3

These conditions mean that $d_{3,2} = \pm 27$; then only these vectors satisfy all four conditions: $(-54,27,27)$, $(54,-27,-27)$, $(0,-27,27)$, and $(0,27,-27)$. But the ones involving ± 54 fail when the constraint for $p = 2$ is considered, as follows. When $d_{3,0} = -54$, B must have exactly 58 multiples of 3. But there are only 31 odd numbers in $[n]$ divisible by 3 and B has only 14 evens by the $p = 2$ work. So if all 31 are in the odd part of B, and all of the 14 evens in B are divisible by 3, the total is $31 + 14 = 45$, short of the needed 58. A similar argument, interchanging A and B, and even and odd, eliminates $d_{3,0} = 54$, and so $\boldsymbol{d}_3$ must be one of $(0, \pm 27, \mp 27)$.

For larger primes, we start with the feasible values of $d_{p,p-1}$, from Proposition 4.2 (1, 3). Then ILP is used to find the possibilities for $d_{p,p-2}$. The binary variables are $x_0, \ldots, x_{183}$. The constraints are those from Propositions 4.2 and 4.3, together with constraints derived from the possibilities for the residue-count vectors (these are easily derived from the difference vectors) from smaller primes. We then continue the same way to find all possibilities for $(d_{p,r}, d_{p,r+1}, \ldots, d_{p,p-1})$. These cases then quickly lead to all the possibilities for $\boldsymbol{d}_p$, using Proposition 4.3 (and the bounds and conditions of Prop. 4.2). When $p = 5$, there are no solutions when $\boldsymbol{d}_3 = (0,27,-27)$.

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

Using $\boldsymbol{d}_3 = (0, -27, 27)$, we find that $\boldsymbol{d}_5$ is $(15, 5, -5, -15, 0)$ or $(5, -15, 15, -5, 0)$. The next case uses the $\boldsymbol{d}_3$ vector and the two for $\boldsymbol{d}_5$ to get $\boldsymbol{d}_7 = (7, -7, 0, 0, 0, 0, 0)$. And then $p = 11$ gives $\boldsymbol{d}_{11} = (-3, -1, 3, -5, 5, -3, 1, 3, 0, 0, 0)$ (and this eliminates the second $\boldsymbol{d}_5$ vector). We next get $\boldsymbol{d}_{13} = (1, -1, 2, -2, 0, 2, 0, 0, 0, -2, 0, 2, -2)$, and when we move to $p = 17$, we find that there are no feasible values of $d_{17,16}$ and hence $184 \notin L_7$.

$n = 168$, $m = 7$. As before, we can deduce that $\boldsymbol{d}_2 = (-64, 64)$, $\boldsymbol{d}_3 = (0, 0, 0)$, $\boldsymbol{d}_5 = (\pm 10, 0, \mp 10, \pm 5, \mp 5)$, and $\boldsymbol{d}_7 = (0, 0, 0, 0, 0, 0, 0)$ (there are 42 other choices for $\boldsymbol{d}_7$ but they are infeasible when the constraints for $\boldsymbol{d}_{11}$ are brought into play). When $p = 11$, we consider two cases according to $\boldsymbol{d}_5$. The first case ($d_{5,1} = +10$), when we use ILP to filter through the constraints for $p = 2, 3, 5$, and 7, leads to 309 possibilities for $\boldsymbol{d}_{11}$. We then try to find a feasible $\boldsymbol{d}_{13}$ vector but there are none. The same works for the second case, with 245 possibilities for $\boldsymbol{d}_{11}$, and so $168 \notin L_7$.

$n = 176$, $m = 8$. Consider first $p = 2$, 3, and 5. For these cases, $r = 0$. A search using the constraints in §4 shows that $\boldsymbol{d}_2 = (0, 0)$, $\boldsymbol{d}_3 = (\pm 27, \pm 27, \mp 54)$ (interchanging A and B if necessary, we may assume $d_{3,0} = 27$), and $\boldsymbol{d}_5$ is one of 16 vectors. These cases lead to residue counts as follows, where the first entry is for residue 0: $p = 2$: $(44, 44)$; $p = 3$: $(43, 43, 2)$, and $p = 5$:

$$(23, 10, 10, 10, 35), (23, 10, 10, 35, 10), (23, 10, 35, 10, 10), (23, 35, 10, 10, 10),$$
$$(18, 5, 5, 30, 30), (18, 5, 30, 5, 30), (18, 5, 30, 30, 5), (18, 30, 5, 5, 30), (18, 30, 5, 30, 5),$$
$$(18, 30, 30, 5, 5), (28, 15, 15, 15, 15), (13, 0, 25, 25, 25), (13, 25, 0, 25, 25),$$
$$(13, 25, 25, 0, 25), (13, 25, 25, 25, 0), (8, 20, 20, 20, 20).$$

All the $p = 5$ cases, except the last, resolve as follows. Consider $(28, 15, 15, 15, 15)$. There are 36 numbers in [176] that are $0 \pmod 5$ and 24 of those are congruent to 0 or 1 mod 3. But we need 28, so there must be at least four that are congruent to 2 mod 3, violating the 2 in $(43, 43, 2)$. This type of counting argument settles the first 11 cases. The next four cases are similar, where one contradicts the 2 by considering two classes with 25 elements each. That leaves only the case of $(8, 20, 20, 20, 20)$.

Using ILP on the constraints for $p = 7$, combined with those for $\boldsymbol{d}_3$ and $\boldsymbol{d}_5$, leads to seven choices for $\boldsymbol{d}_7$. Moving to $p = 11$, Proposition 4.2 yields $17^3 = 4913$ possibilities for $(d_{11,8}, d_{11,9}, d_{11,10})$ and when each of these is extended to an 11-vector by Proposition 4.3(2) and then filtered through the choices for $\boldsymbol{d}_3$ and $\boldsymbol{d}_5$, only 208 possibilities remain. Bringing in $\boldsymbol{d}_7$ reduces these to 23 possibilities, and then $p = 13$ reduces these to only the balanced case: $\boldsymbol{d}_{11} = (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$. Having a unique $\boldsymbol{d}_{11}$ allows $p = 2, 3, 5$, and 11 to reduce the $14^5 = 537624$ possible terminal 5-tuples of $\boldsymbol{d}_{13}$ down to 977 cases. Then filtering through $\boldsymbol{d}_7$ reduces these to just 31 cases, and they all resolve negatively when $p = 17$ is used together with the constraints for $p = 2, 3, 5, 7, 11$, and 13. This proves that $176 \notin L_8$.

$n = 272$, $m = 9$. This case is similar to the others. The usual constraint analysis, filtering through smaller primes as we move up, gives $\boldsymbol{d}_2 = (0, 0)$, $\boldsymbol{d}_3 = (81, -81, 0)$, $\boldsymbol{d}_5 = (25, -25, 0, 0, 0)$ and only these four feasible cases of $(\boldsymbol{d}_7, \boldsymbol{d}_{11}, \boldsymbol{d}_{13})$:

$$(7, 7, -7, 7, -7, -7, 0), (-7, -5, -3, -1, 1, 3, 5, 7, -2, 0, 2),$$
$$(1, -5, 1, -1, -5, -5, 5, 5, 1, -1, 5, -1, 0)$$

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

$$(7,7,-7,7,-7,-7,0),(-7,-5,-3,-1,1,3,5,7,-2,0,2),$$
$$(-1,5,-1,1,5,5,-5,-5,-1,1,-5,1,0)$$

$$(-7,-7,7,-7,7,7,0),(-3,1,5,9,-9,-5,-1,3,-4,0,4),$$
$$(1,-5,1,-1,-5,-5,5,5,1,-1,5,-1,0)$$

$$(-7,-7,7,-7,7,7,0),(7,5,3,1,-1,-3,-5,-7,2,0,-2),$$
$$(-5,-1,-5,5,-1,-1,1,1,-5,5,1,5,0).$$

These are quickly shown to be infeasible when the constraints for $\boldsymbol{d}_{17}$ are used, proving $272 \notin L_9$. □

The positive results regarding L_m (§6) can be verified in an instant. Most of the negative results can be obtained essentially by hand. When $n = 168, 176, 184$, and 272 we used ILP, which cannot be certified correct. But we verified the results in different software environments with somewhat different algorithms. The first three cases were proved in 2007 by the second author (using lp_solve and requiring a few days on a cluster of 100 computers) and we repeated that work using *Mathematica*; the latter took about two days of computation time and succeeded without the need for any constraints derived from power-sum equalities. The intermediate data was consistent with the 2007 work. For $n = 272$, the intermediate results were computed using both lp_solve and *Mathematica* and the lists of vectors agreed. For ILP work, the second author uses lp_solve [3], the third author uses *SAS* [15], and the fourth author uses *Mathematica* [19], which calls on COIN-OR for ILP. Comparisons of the three solvers can be found in [13].

Assuming the Symmetry Conjecture given below, and using the techniques of Theorem 5.1 augmented by the extensions for d_4 and d_9 in [2], we showed that the smallest possible entry listed for $m^*(n)$ in Table 2 is the correct value for $n = 240, 288,$ and 304.

6. Existence of equipowerful bipartitions

Here we will find equipowerful sets that prove Theorems 3.1, 3.2, and 3.3. Symmetry plays a key role even in Theorem 3.1, where it is not explicitly mentioned. All of our computational evidence supports the following idea.

Symmetry Conjecture. *If $\mathcal{P}(n,m)$ is nonempty then it contains an LP f that is $(-1)^m$ symmetric. Alternatively, if there is an equipowerful bipartition of length n and exact order m then there is a witnessing set that is $(-1)^m$ symmetric.*

For instance, the 3-equipowerful set $\{0,2,6,7,8,10\} \subset [12]$ given in §4 is anti-symmetric.

This idea has two important implications: (1) Searches should be streamlined (i.e., made feasible) by just assuming $(-1)^m$ symmetry, as needed. (2) The Symmetry Conjecture should be tested in all situations where we know that $\mathcal{P}(n,m)$ is nonempty, and testing is feasible.

In fact, many of our searches would not have been possible without making the assumption in (1). Moreover, we were able to succeed for all of the (infinitely many) n implicitly asserted in Table 1 by using the regenerative pairs to be described shortly.

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

The characterizations for $m \le 6$ (Table 1) were all known, but the symmetry aspect of Theorem 3.2 is new for $3 \le m \le 8$. To prove Theorem 3.1 (as well as the known characterizations for $m \le 6$), one can start with the trivial $m = 0$ case and use Doubling in Theorem 4.1 to move up, while finding enough additional examples so that Addition in Theorem 4.1 leads to the infinite family of witnessing sets. For example, when $m = 2$, the base case is $\{0, 3\} \subset [4]$ and Addition handles the rest. When $m = 3$, doubling the $m = 2$ case gives $8 + 8\mathbb{N}$, leaving $12 + 8\mathbb{N}$ unresolved. But once we have the $n = 12$ witness, Addition $(12 + 8,\ 12 + 8 + 8,\ \dots)$ takes care of the rest. This method, once other sporadic examples are taken care of, handles all $m \le 8$. Since Addition does not preserve symmetry, this method will not yield sets having the desired symmetry properties.

TABLE 3. Sporadic cases for symmetry.

m	n
3	12
4	24
5	40, 56
6	48, 72, 88, 104
7	112, 200, 216, 232, 248, 264, 280, 296, 312, 328
8	144, 208, 240, 272, 304
9	192, 240

We found appropriately symmetric sets for all the needed sporadic cases, which are listed in Table 3. The largest example shows $328 \in L_7$ and is the antisymmetrization of

$$\{0, 3, 5, 7, 9, 11, 12, 15, 16, 18, 19, 21, 23, 27, 29, 30, 33, 35, 37, 39, 41, 42, 45, 47,$$
$$49, 51, 53, 55, 57, 59, 62, 63, 65, 66, 67, 69, 71, 73, 75, 77, 81, 83, 84, 87, 89, 91, 93, 95,$$
$$97, 99, 100, 102, 106, 107, 111, 112, 113, 114, 116, 117, 119, 122, 125, 128, 129, 130,$$
$$131, 133, 134, 137, 139, 143, 144, 146, 149, 150, 152, 153, 160, 161, 162, 163\}.$$

The sum of the 6th powers of this set is 28863168757954570, which equals the 6th power sum for its complement in [328]; this power-sum equality holds for all exponents up to 6. The corresponding LP factors into the product of $(1 - x)^7$ and an irreducible polynomial of degree 320. The witnesses for all 25 needed cases follow from the data in Tables 4 and 5.

The raw search space for the 328 example has more than 10^{97} sets, so clearly some tricks are needed to get the search to work. The main tool is integer linear programming: a binary variable is used for each value in $[n]$ and the power-sum constraints are then linear equations. In addition we use the following constraints and tricks.

1. We assume the Symmetry Conjecture, which halves the variable count. By Symmetry in Theorem 4.1, this means that the power constraints need only go to the $(m - 2)$nd power, as the last one comes for free.

2. We use the modular constraints for small primes derived from the results in §4. Further, we filter the constraints down to ones that are consistent with the assumed symmetry property. Recall that the constraints are first derived for the vectors $\boldsymbol{d}_p$; they are then used to get the counts for the congruence classes in the

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

set A. To filter the set of $\boldsymbol{d}_p$ as needed for $(-1)^m$ symmetry, keep only those $\boldsymbol{d}_p$ for which $d_{p,i} = (-1)^m d_{p,n-i-1}$ where $0 \le i \le p-1$ and $n-i+1$ is reduced modulo p.

3. To avoid problems with the very large numbers that arise, we shift the domain from $[n]$ to the interval $\left[-\frac{n}{2}+1, \frac{n}{2}\right]$. This is allowed because if $A \stackrel{m}{=} B$ as subsets of $[n]$, then $A + t \stackrel{m}{=} B + t$, a fact that is easily proved by using the Equivalence Lemma and observing that multiplication by x^t does not affect the power of $1-x$ that divides a polynomial.

4. For the same reason as in (3), we replace powers by binomial coefficients. The definition of an m-equipowerful number uses powers x^j, but any family of m polynomials of degree $0, 1, \ldots, m-1$ that takes integer values on integer arguments can be used instead. If the ILP search uses as its main constraint not power identities but instead equalities for the sums of the binomial coefficient polynomials over A, B then the size of the numbers is substantially reduced. Consider the search to show $240 \in L_9$; using powers (and also points 1 and 3 above) involves numbers as large as 120^8, about $4 \cdot 10^{16}$, compared to the binomial coefficient $\binom{120}{8}$, which is about $8 \cdot 10^{11}$.

Addition, in Theorem 4.1, fails to preserve symmetry so a proof of Theorem 3.2 requires a new type of rule that does. This is what the concept of *regenerative pairs* accomplishes. We stumbled on this idea when we realized that the Symmetry Conjecture had not been proved even for the case $m = 3$. To handle that case we needed a new way to go from 12 to $20, 28, \ldots$. We found a way to do this and then found several other instances where $f \in \mathcal{P}(n, m)$ could be extended to an LP in $\mathcal{P}(n + n', m)$ so that symmetry is preserved. In short, we found a new type of "addition rule" that respects symmetry.

Theorem 6.1. *Fix an integer m and let $s = (-1)^m$. For any LP h, let $S(h) = h \vee sh^*$ be the s-symmetrization of h. Assume that f and g are LPs of lengths n and δ, respectively. Define LPs*

$$f_1 = f \vee g, \qquad f_2 = f \vee g \vee (-1)^m g^* = f \vee S(g)$$

of lengths $n + \delta$ and $n + 2\,\delta$, respectively. Then if $S(f)$ and $S(f_1)$ are each divisible by $(x-1)^m$, then so is $S(f_2)$. In other words:

$$\text{if } \; S(f) \in \mathcal{P}(2n, m) \;\text{ and }\; S(f_1) \in \mathcal{P}(2n + 2\delta), \;\text{ then }\; S(f_2) \in \mathcal{P}(2n + 4\delta, m).$$

In words: if an LP f_1 extends an LP f, and both $S(f)$ and $S(f_1)$ have order m, then there is a closely related f_2, extending f_1, such that $S(f_2)$ has order m. Continuing in this way gives an infinite sequence of LPs of order m whose lengths form an arithmetic progression.

Definition 6.2. If f, of length n, and $f_1 = f \vee g$, of length $n_1 = n + \delta$, satisfy the hypotheses of Theorem 6.1, then (f, f_1) is a *Regenerative Pair* (RP) for $(2n, 2n_1)$.

Remark. The idea of joining an LP to a reflection (possibly negated) has been in the LP literature from the beginning [2, 11], but the more specific nature of an RP allows for a useful dance that gives lengths lying in an arithmetic progression.

Examples.

1. The LPs $1-x+x^2-x^3-x^4-x^5$ and $1-x+x^2-x^3-x^4-x^5+x^6+x^7+x^8+x^9$ are an RP for $(12, 20)$: the antisymmetrizations of the polynomials are in $\mathcal{P}(12, 3)$, $\mathcal{P}(20, 3)$, respectively.

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

2. If f is such that $S(f) \in \mathcal{P}(2n, m)$, where S is $(-1)^m$ symmetrization, then (f, f_1) is an RP for $(2n, 4n)$ where f_1 is the length-$2n$ initial part of $S(f) \vee -S(f)$.
3. For all m, the pair (τ_m, τ_{m+1}) is an RP for $(2^{m+1}, 2^{m+2})$.
4. *Doubling* an RP. It is easy to prove that if $(f, f \vee g)$ is an RP for (n_0, n_1) of order m then $((1-x)\#f, (1-x)\#(f \vee g))$ is an RP for $(2n_0, 2n_1)$ for $m+1$.

Proof of Theorem 6.1. Throughout this proof $\equiv$ denotes congruence modulo $(x-1)^m$. Let $n_1 = n + \delta$ and $n_2 = n + 2\,\delta$ be the lengths of f_1 and f_2. From the hypotheses, we have

$$S(f) = f + sx^n f^* \equiv 0$$

$$S(f_1) = f_1 + sx^{n+\delta} f_1^* = f + x^n g + s\,x^{n+\delta} g^* + s\,x^{n+2\delta} f^* \equiv 0.$$

Subtracting the first congruence for $S(f)$ from the one for $S(f_1)$, and dividing by x^n, gives

$$-s\,f^* + g + s\,x^\delta g^* + s\,x^{2\delta} f^* \equiv 0.$$

Moving the f^* terms to the right side gives $S(g) \equiv s\,(1 - x^{2\,\delta}) f^*$. Multiplying by $1 + x^{2\delta}$ then gives

$$\left(1 + x^{2\,\delta}\right) S(g) \equiv s\,(1 - x^{4\,\delta}) f^*.$$

Multiply by x^n, replace $s\,x^n\,f^*$ by $-f$, bring everything to the left side, and expand to get

$$f + x^n g + sx^{n+\delta} g^* + x^{n+2\delta} g + s\,x^{n+3\delta} g^* + s\,x^{n+4\,\delta} f^* \equiv 0.$$

This can be carefully checked, from the definition of f_2, to give the desired conclusion: $S(f_2) \equiv 0$. □

The power of this theorem is that (f_1, f_2) becomes an RP, with lengths n_1 and $n_1 + \delta$, so the theorem can be applied again. We can iterate forever, concluding that L_m contains $2n, 2n + 2\delta, 2n + 4\delta, 2n + 6\delta, \ldots$. Moreover, all the witnesses will be $(-1)^m$ symmetric.

Corollary 6.3. *With notation as in Theorem* 6.1*, for any $j \geq 0$, define*

$$G_j = \bigvee_{i=1}^{j} h_i$$

where $h_i = g$ if i is odd and $h_i = s\,g^*$ if i is even. Then $S(f \vee G_j)$ is in $\mathcal{P}(2n+2j\delta, m)$ and is $(-1)^m$ symmetric.

Continuing with the $(12, 20)$ example, the antisymmetrization of f corresponds to $\{0, 2, 6, 7, 8, 10\}$, while the same for f_1 gives $\{0, 2, 6, 7, 8, 9, 14, 15, 16, 18\}$. These two sets are antisymmetric witnesses to $12, 20 \in L_3$. Because the left halves of these sets are nested, the iterative construction of Theorem 6.1 leads to the single infinite set $X = \{0, 2, 6, 7, 8, 9, 14, 15, 16, 17, 22, 23, 24, 25, \ldots\}$. This single set provides antisymmetric witnesses for $12 + 8\mathbb{N}$: just take the appropriate initial segment and antisymmetrize it. The difference sequence of X (assuming 0 is in the set) is ultimately periodic: $2, 4, \overline{1, 1, 1, 5}$, where the bar indicates repetition. This finitary method of witnessing infinitely many numbers in L_m will occur whenever we have an RP: there will be a single ultimately periodic difference sequence, which defines an infinite set X that is a union of finitely many arithmetic progressions. Using $(-1)^m$ symmetrization on appropriate initial segments of X will yield $(-1)^m$ symmetric witnesses for infinitely many values in L_m.

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

So we can prove Theorems 3.2 and 3.3 by finding RPs for the needed cases. It took several weeks, but the ILP method with the various constraints succeeded in finding all the required pairs. For each $m \leq 6$ the asymptotic result follows from a single RP (Table 4). But it took nine RPs to cover L_7. Doubling the $m = 7$ case covers almost all of L_8, and the characterization of that case is completed by finding seven additional 8-equipowerful sets, shown in Table 5.

In order to compactly represent all of our results, we will use hexadecimal notation in the following way. Convert the characteristic function of $A \subseteq [n]$ into a binary string, and convert that string to hex. Due to symmetry or antisymmetry, we need only provide the left half.

For example, the RP for $(12, 20)$ is defined by $\{0, 2, 6, 7, 8, 9, 14, 15, 16, 18\} \subset [20]$. The corresponding bit-string is 1010 0011 1100 0011 1010, which is A3C3A in hexadecimal. The leftmost three digits encode the required half of the set, so the set is given by A3C.

Table 4 shows all the needed RPs for the asymptotic results (the $m = 8$ case follows by just doubling the sets from $m = 7$); the first few rows also contain the sets explicitly. Table 5 shows individual examples for the cases not covered by the pairs. In all cases only the left half of the sets is encoded, as the full set is obtainable by $(-1)^m$ symmetrization.

TABLE 4. RPs that, with only few exceptions, cover all lengths $n \in L_m$, for $m \leq 8$.

m	lengths	hex code for regenerative pair
3	$12, 20$	A3C $\{0, 2, 6, 7, 8, 9\}$
4	$24, 32$	995C $\{0, 3, 4, 7, 9, 11, 12, 13\}$
5	$48, 64$	apply Doubling to the example for $m = 4$
5	$56, 72$	96A371999 $\{0\,3\,5, 6, 8, 10, 14, 15, 17, 18, 19, 23, 24, 27, 28, 31, 32, 35\}$
6	$104, 120$	C32F8696687E155
6	$112, 128$	936D342C73B58A0F
7	$208, 224$	A55A9936CCA5363A65E41B6CC35A
7	$200, 328$	9559B5165565555553755459555A31ED24F651A6C0F
7	$216, 344$	953D547143575D547554D551077417ED3C9A6982C37
7	$232, 360$	95555D95510D77154D75994531D714F238C7EB8831D1D
7	$248, 376$	C557185D75534395571D14715DC7534EC412F9C933CB199
7	$264, 392$	D29535559352D575954B549555A54B555C968936B1BDC6538
7	$280, 408$	A8ADA3CAE2B88E2B9AC32CAE2B8EE2B8AC347D930FE4826A3C7
7	$296, 424$	9996356B5496555555555E925945DA7532555477C255A9AA6587A
7	$312, 440$	A56A69555555B59550965AD5B71515554E56725A64EA946FB10D563

The method of proof using RPs leads to a surprising amount of structure in L_m when $m \leq 8$. Consider the trivial fact that, for L_2, the symmetry result can be proved by a single set. Just let $X = \mathbb{N}$ and get the witness for $4\,k$ by taking the first k entries in X and symmetrizing the result. For L_3, we have two RPs that yield antisymmetric witnesses for all cases: the 12/20 case and the 8/16 case. For L_4 a single RP takes care of $24 + 8\mathbb{N}$ and that leaves only the singleton 16, which, by the example following Definition 6.2, can be viewed as the first half of an RP. If we ignore the small number of exceptions, we see from Table 4 that up to $m = 6$ we have at most two RPs that cover L_m. At $m = 7$ we need nine: eight to cover the numbers that do not arise by doubling L_6, and one more (208/224) to cover those

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

TABLE 5. Sporadic examples.

m	n	hex code for witness to $n \in L_m$
5	40	C1EC9 $\{0,1,7,8,9,10,12,13,16,19\}$
6	48	C27D8C $\{0,1,6,9,10,11,12,13,15,16,20,21\}$
6	72	998EAA3C5 $\{0,3,4,7,8,12,13,14,16,18,20,22,26,27,28,29,33,35\}$
6	88	96C362DE4A3
7	112	A5994DB29B45A0
8	144	A962D5AF05357231CD
8	208	C5A2CB6B945B28BC43ADB6586C
8	272	D0EC315B66B4DE6110D339D63B4CD61BC4
8	304	A9E21ACCB5C96794C56D9C4255B4336DEAD8A4
8	336	92D6F430CB2AD476994F60DEB609EA1158973CA7D2
8	368	B4D121ABDED920EAF02CE76086B55BA6995ABE9A43247A
9	192	C1BE1E21CD63D295A7887A59
9	240	9666995C93C3AA5935E81CB7C2938D
10	240	C394E65AB989457A86B7D0D921636E

that do arise by such doubling. The case $m = 8$ follows by Doubling the examples for $m = 7$. One can ask whether a finite covering set of RPs always exists. As pointed out after Corollary 6.3, this structure means that there are finitely many sets X_i, each of which is a union of arithmetic progressions, so that the appropriate symmetrization of initial segments of the X_i leads to $(-1)^m$ symmetric witnesses for all $n \in L_m$, with finitely many exceptions. However, by Example 2 after Definition 6.2, any LP is the beginning of a trivial RP, so that perhaps these aren't truly exceptions.

7. Conclusion and questions

It is easy to make conjectures based on the patterns observed in data. This area is remarkable for the number of such guesses that have turned out to be wrong.

For instance, looking at L_m for $m \le 5$ suggests the natural idea that the Thue–Morse polynomial τ_m will be the order-m LP of smallest length, so that $2^m = \min L_m$. This was disproved by Boyd [6] and Skachek [17] who found $48 \in L_6$. Another conjecture arising from $m \le 5$ is that τ_m is the unique (up to sign) LP in $\mathcal{P}(2^m, m)$; this fails because, again by Boyd, $\mathcal{P}(64, 6)$ has three LPs (up to sign). Increasing m, one sees that for $m \le 9$, $\min L_m > 2^{m-1}$, and one might be tempted to guess that this is always true. Although nothing in our data immediately contradicts this, note that if $f \in \mathcal{P}(144, 8)$, then by Multiplication in Theorem 4.1, $f \# f \in \mathcal{P}(144^2, 16)$ and so $\min(L_{16}) \le 144^2 < 2^{15}$. However, from all of the evidence from our tables it still may seem natural to conjecture that 2^{m-1} is never in L_m. Richard Stong [18] found a clever way to combine known elements of various $\mathcal{P}(n, m)$ to disprove this; here, with his kind permission, is his result.

Theorem 7.1. $\mathcal{P}(2^{51}, 52) \neq \varnothing$.

Proof. Let $p_{n,m}$ be any element of $\mathcal{P}(n, m)$ and let $p^{\#j}$ denote $p \# p \# \cdots \# p$, with j terms. Recalling that $p_{n,m} \# p_{n',m'} \in \mathcal{P}(nn', m + m')$, we define four large (in fact,

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

gigantic compared to anything discussed earlier) LPs, and their join:

$$\begin{aligned}
a &= p_{144,8}^{\#2} \# p_{192,9}^{\#4} && \in \mathcal{P}(144^2 \cdot 192^4, 16+36)\\
b &= p_{48,6}^{\#3} \# p_{112,7} \# p_{192,9}^{\#3} && \in \mathcal{P}(48^3 \cdot 112 \cdot 192^3, 18+7+27)\\
c &= p_{16,4} \# p_{112,7}^{\#2} \# p_{208,8} \# p_{272,8} \# p_{192,9}^{\#2} && \in \mathcal{P}(16 \cdot 112^2 \cdot 208 \cdot 272 \cdot 192^2,\\
& && \qquad 4+14+8+8+18)\\
d &= p_{112,7} \# p_{208,8}^{\#4} \# p_{8192,13} && \in \mathcal{P}(112 \cdot 208^4 \cdot 8192, 7+32+13)\\
f &= a \vee b \vee c \vee d.
\end{aligned}$$

From $p_{n,m} \vee p_{n',m} \in \mathcal{P}(n+n', m)$, the fact that all ms in the definitions of a, b, c, and d are equal to 52, and the surprising arithmetic fact

$$\begin{aligned}
&144^2 \cdot 192^4 + 48^3 \cdot 112 \cdot 192^3 + 16 \cdot 112^2 \cdot 208 \cdot 272 \cdot 192^2 + 112 \cdot 208^4 \cdot 8192\\
&= 28179280429056 + 87668872445952 + 418591807635456 + 1717359853174784\\
&= 2^{51}
\end{aligned}$$

it follows that $f \in P(2^{51}, 52)$ as desired, and $2^{51} \in L_{52}$. □

The arithmetic fact was found by compiling a useful list of rational numbers $n/2^m$ for which $\mathcal{P}(n,m) \neq \varnothing$, and then doing a search (in *Mathematica*) for a subset that summed to $1/2$.

Recall that the Thue–Morse polynomial τ_m is in $\mathcal{P}(2^m, m)$. Define the "mth error" of $f \in \mathcal{P}(n,m)$ to be the absolute value of $f^{(m)}(1)$. Then we have

$$f^{(m)}(1) = \sum a_i\, i(i-1)\ldots(i-m+1) = \sum a_i\, i^m$$

where the last sum is a moment of the corresponding set bipartition and the last equality holds because all the smaller moments (and corresponding derivatives) are 0. The mth error of τ_m is easily seen to be $(-1)^m 2^{m(m-1)/2} m!$.

In [1, Open Problem 6.12.5] Allouche and Shallit ask whether τ_m is the element of $\mathcal{P}(2^m, m)$ having the smallest mth error. For example, $\mathcal{P}(2^6, 6)$ has two LPs in addition to τ_6 and the three 6th errors are approximately 24 million for τ_6, and 93 million and 350 million for the other two.

Fix $m = 51$. Then the f in Theorem 7.1 has m-th error equal to 0, giving a remarkably definitive answer to their question. More precisely, f and τ_{51} both lie in $\mathcal{P}(2^m, m)$, and f has m-th error equal to 0 since it is actually divisible by $(x-1)^{52} = (x-1)^{m+1}$.

Given the sobering record of failed guesses and conjectures discussed above, caution is in order. So we end with a list of questions, and are only willing to label the first as a conjecture.

- (Symmetry Conjecture) If $\mathcal{P}(n,m) \neq \varnothing$, then it contains a $(-1)^m$ symmetric LP. (For $m \leq 8$, this holds by Theorem 3.2.)
- Is it the case that, for each m, there is a finite family of RPs that provides $(-1)^m$ symmetric witnesses for each entry in L_m? (For $m \leq 8$, this is true by the proof of Theorem 3.2.)
- If $n = \min L_m$, is it true that all $f \in \mathcal{P}(n,m)$ having exact order m are $(-1)^m$ symmetric? (This is the case for $m \leq 6$.)
- What is the smallest k so that $k + 16\mathbb{N} \subset L_9$?

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.

- What is the asymptotic behavior of $\min L_m$? We know that
$$\exp((1+o(1))\sqrt{m}) \le \min L_m \le O(c^m), \quad \text{for } c = 192^{1/9} \sim 1.79349\ldots$$
where the upper bound can be obtained by using expanded products of the element of $\mathcal{P}(192, 9)$.

References

[1] J.-P. Allouche and J. Shallit, *Automatic Sequences*: *Theory, applications, generalizations*, Cambridge University Press, Cambridge, 2003. MR1997038

[2] D. Berend and S. Golan, *Littlewood polynomials with high order zeros*, Math. Comp. **75** (2006), no. 255, 1541–1552, DOI 10.1090/S0025-5718-06-01848-5. MR2219044

[3] M. Berkelaar, K. Eikland, P. Notebaert, et al., *lpsolve: Open source (mixed-integer) linear programming system*, Eindhoven U. of Technology **63** (2004).

[4] P. Borwein and C. Ingalls, *The Prouhet-Tarry-Escott problem revisited*, Enseign. Math. (2) **40** (1994), no. 1-2, 3–27. MR1279058

[5] P. Borwein and M. J. Mossinghoff, *Polynomials with height 1 and prescribed vanishing at 1*, Experiment. Math. **9** (2000), no. 3, 425–433. MR1795875

[6] D. W. Boyd, *On a problem of Byrnes concerning polynomials with restricted coefficients*, Math. Comp. **66** (1997), no. 220, 1697–1703, DOI 10.1090/S0025-5718-97-00892-2. MR1433263

[7] D. W. Boyd, *On a problem of Byrnes concerning polynomials with restricted coefficients. II*, Math. Comp. **71** (2002), no. 239, 1205–1217, DOI 10.1090/S0025-5718-01-01360-6. MR1898751

[8] H. L. Dorwart and O. E. Brown, *The Tarry-Escott Problem*, Amer. Math. Monthly **44** (1937), no. 10, 613–626, DOI 10.2307/2301480. MR1524120

[9] G. Freiman and S. Litsyn, *Asymptotically exact bounds on the size of high-order spectral-null codes*, IEEE Trans. Inform. Theory **45** (1999), no. 6, 1798–1807, DOI 10.1109/18.782100. MR1720633

[10] S. Golan, *Equal moments division of a set*, Math. Comp. **77** (2008), no. 263, 1695–1712, DOI 10.1090/S0025-5718-08-02072-3. MR2398788

[11] D. Katz, S. Lee, and S. Trunov, *Rudin-Shapiro-like sequences with maximum asymptotic merit factor*. `arXiv:1711.02233v4` 2020.

[12] R. Lidl and H. Niederreiter, *Finite Fields*: *With a foreword by P. M. Cohn*, 2nd ed., Encyclopedia of Mathematics and its Applications, vol. 20, Cambridge University Press, Cambridge, 1997. MR1429394

[13] H. Mittelmann, *Decision tree for optimization software*. `http://plato.asu.edu/bench.html/`.

[14] R. M. Roth, P. H. Siegel, and A. Vardy, *High-order spectral-null codes: constructions and bounds*, IEEE Trans. Info. Th. **40** (1994), 1826–1840.

[15] SAS Institute Inc., *SAS Optimization 8.5*, SAS Institute Inc., Cary, NC, 2019.

[16] K. Schouhamer Immink and K. Cai, *Estimated spectra of higher-order spectral null codes*, IEEE Communications Letters **23** (2019), 20–23.

[17] V. Skachek, *Coding for Spectral-null Constraints*, M. Sc. thesis (in Hebrew), Technion, 1997.

[18] R. Stong, personal communication, 2019.

[19] Wolfram Research, Inc. *Mathematica*, version 12.1.1, Champaign, IL.

Department of Mathematics, University of Minnesota, Minneapolis, Minnesota 55455
Email address: `jbuhler@math.umn.edu`

Department of Computer Science, Jerusalem College of Technology, Jerusalem, Israel
Email address: `sgolan@jct.ac.il`

SAS Institute, Inc., Cary, North Carolina 27513
Email address: `rob.pratt@sas.com`

Department of Mathematics, Macalester College, St. Paul, Minnesota 55105
Email address: `wagon@macalester.edu`

This is a free offprint provided to the author by the publisher. Copyright restrictions may apply.